

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit:** Department of Geology and Geophysics

2. **Subject Name:** Hydrogeology Lab **L-T-P:** 0-0-3 **Credits:** 2

3. **Pre-requisites:**

4. **Syllabus and reference books:**

Syllabus:

Laboratory component to explain the basic concepts of groundwater science and its relation with Earth systems; its impact on human and society; groundwater as a part of hydrosphere, hydrological equations, modeling of the hydrologic cycle, the water profile, sources of water, surface-groundwater interactions, recharge, and discharge processes, hydraulic parameters, and their influence on hydro dynamics groundwater flow, hydrogeochemistry, groundwater quality, types of groundwater, sources and pathways of groundwater solutes

Reference Books:

1) Domenico, P.A. and Schwartz, F.W., 1998. *Physical and chemical hydrogeology*. New York: Wiley.

2) Fetter, C.W., 2018. *Applied hydrogeology*. Waveland Press

1) Tóth, J., 2009. *Gravitational systems of groundwater flow: theory, evaluation, utilization*. Cambridge University Press

2) Mukherjee, A., Scanlon, B.R., Aureli, A., Langan, S., Guo, H. and McKenzie, A.A. eds., 2020. *Global groundwater: source, scarcity, sustainability, security, and solutions*. Elsevier

5. **Lecture-wise break-up:**

Sl. No.	Topic	No. of lectures
1.	Application of Hydrologic equations	6
2.	Concepts of fluid pressure, potentiometric maps, groundwater flow and interaction with surface water	6
3.	Mapping of groundwater flow using analytical and numerical methods	6
4.	Well hydraulics	6
5.	Fieldwork	6
6.	Groundwater Chemistry	6
Total number of hours		36